Name- Vanshika Dixit Roll number - 2401010010 B-Tech Cse [Core]

12/2/26

[DBMS assignment - 2]

[Section - A]

1. Entity Relationship (ER) Diagram
2. Relational model
3. Object - oriented model
4. Integrity Constraints

[Section - B]

Q.2 Data abstraction means hiding complex database details and showing only necessary information to users.

In a healthcare system, data abstraction helps doctors, nurses and administrators access patient information easily without dealing with complex database structures.

How it simplifies interaction

1. Simplified interface
2. Role-based access
3. Improved efficiency
4. Data Security
5. Reduced Complexity

Thus data abstraction makes patient record management efficient, secure and easy to use.

Q.3 Benefits

1. Ease of use - Healthcare professionals can access information without technical database knowledge.
2. Improved data security - Sensitive patient data is protected by restricting access.
3. Data Consistency - Abstraction ensures that all users view consistent and accurate information.
4. Better system maintenance - Database changes can be made internally without affecting users.
5. Improved efficiency in healthcare operations - Quick access to patient records helps doctors make faster decisions.

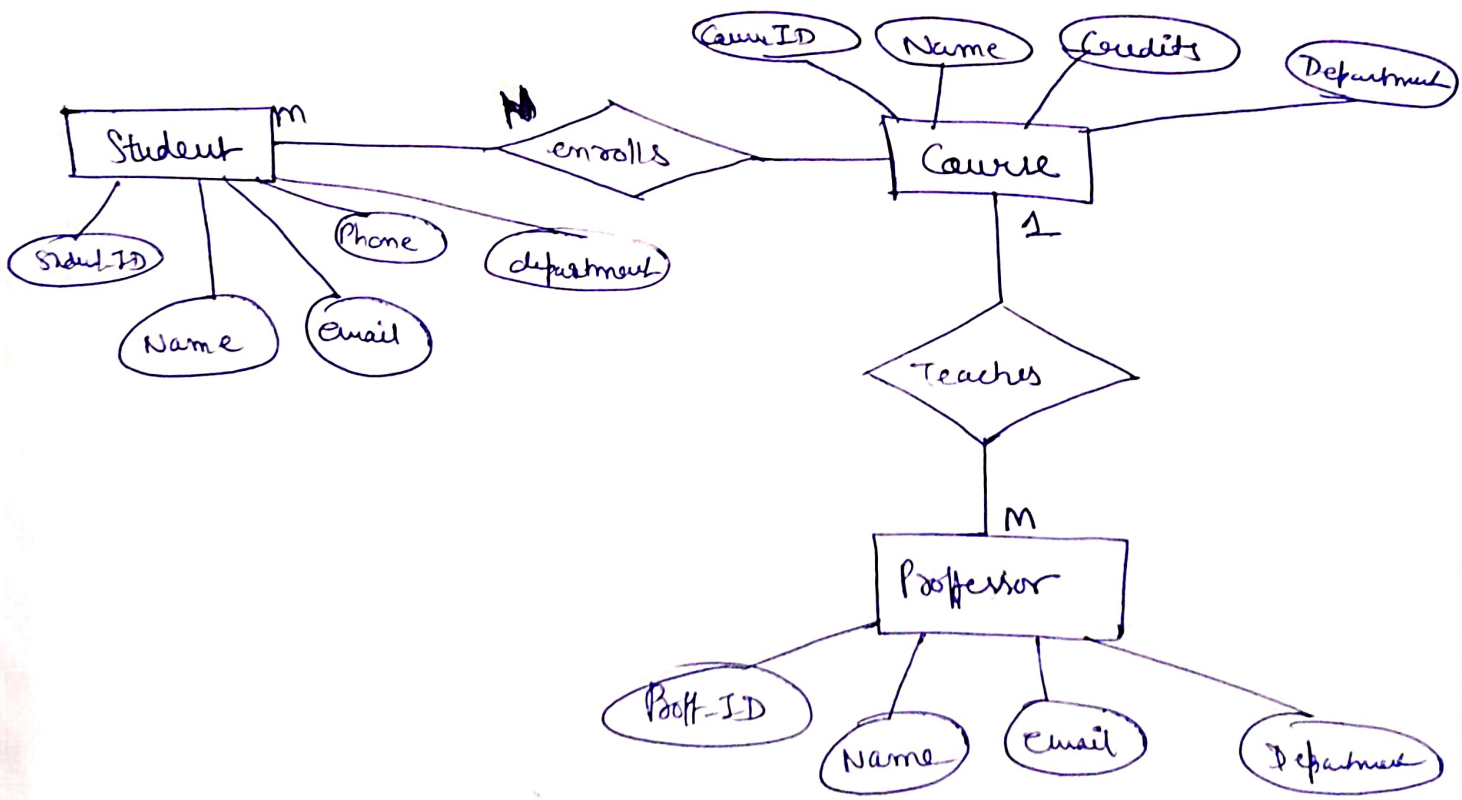
Challenges

1). Implementation Complexity
2). Performance overhead
3). High development cost
4). Security risks if poorly implemented
5). Training requirements

[Section-C]

Student (Student_ID (PK), Name, Email, Phone, Department)
Course (Course_ID (PK), Name, Credits, Department)
Attributes (Proff_ID (PK), Name, Department)

⇒ Relationships

1. Student — Enrolls — Course
m : m

2. Proff — Teaches — Course
1 : m